MNIST HANDWRITTEN DIGIT RECOGNITION

Deep Learning Mini Project Report

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Abstract

This project demonstrates handwritten digit recognition using the MNIST dataset and a neural-network-based deep-learning model. MNIST contains grayscale images of handwritten digits from 0 to 9. The notebook loads and preprocesses the data, trains a model, evaluates performance, and saves the trained model for later use.

1. Introduction

Handwritten digit recognition is an image-classification task in which a computer predicts the number shown in an image. It is a useful introductory project for learning data preparation, neural networks, model evaluation, and saving a trained model.

2. Problem Statement

Develop a program that classifies an image of a handwritten digit into one of ten classes: 0 through 9.

3. Objectives

- Understand a basic deep-learning workflow.
- Load and preprocess handwritten digit images.
- Train a neural network to classify digits.
- Evaluate the model on test data.
- Save the trained model for future predictions.

4. Dataset: MNIST

MNIST is a widely used handwritten-digit dataset. Each image is a 28 × 28 pixel grayscale image and has a label from 0 to 9. The standard dataset contains 60,000 training images and 10,000 test images. Pixel values are commonly normalized from 0–255 to 0–1.

5. Tools and Technologies

- Python — programming language.
- Google Colab / Jupyter Notebook — execution environment.
- TensorFlow / Keras — model building and training, if used in the notebook.
- NumPy and Matplotlib — numerical operations and plots, if used.
- GitHub — project storage and sharing.

6. Methodology

1. **Load data:** Load MNIST images and their correct labels.
2. **Preprocess:** Normalize pixel values and prepare input dimensions.
3. **Build model:** Define a neural-network architecture for digit classification.
4. **Train:** Fit the model using training images and labels.
5. **Evaluate:** Measure performance on test data.
6. **Save:** Save the trained model as models/mnist_digit_model.keras.

7. System Workflow

Input image → preprocessing → trained neural network → predicted digit (0–9) → evaluation against the correct label.

8. Results and Output

The notebook displayed the completion message: “Saved model to models/mnist_digit_model.keras. Project finished. Check outputs/ for graphs and screenshots.” Add the exact test accuracy printed by your notebook and include screenshots of training results or predictions before submission.

Test accuracy reported by notebook: _____

Screenshot / sample prediction reference: _____

9. Applications

- Reading handwritten numbers on forms.
- Digit recognition in document processing.
- Educational demonstrations of image classification.
- A foundation for advanced computer-vision projects.

10. Limitations

A model trained on MNIST may not perform equally well on photos, colored images, or handwriting with different backgrounds. Input images need correct resizing and preprocessing.

11. Future Enhancements

- Create a web interface where users can draw digits.
- Add a confusion matrix and detailed metrics.
- Test on user-drawn images with suitable preprocessing.
- Compare a basic neural network with a convolutional neural network (CNN).

12. Conclusion

The project demonstrates an introductory supervised deep-learning workflow, including data preparation, training, evaluation, and model saving. It provides a foundation for learning neural networks and image classification. Add the actual accuracy and screenshots from your own notebook before final submission.

References

- MNIST database: http://yann.lecun.com/exdb/mnist/
- TensorFlow: https://www.tensorflow.org/
- Google Colab: https://colab.research.google.com/
- GitHub Docs: https://docs.github.com/